USD cash reserves

Cash timing and purchasing power over short holding periods

A supervised AI dashboard for a family-office client

Group g00: development code, replace before submission

Educational project. Not investment advice.

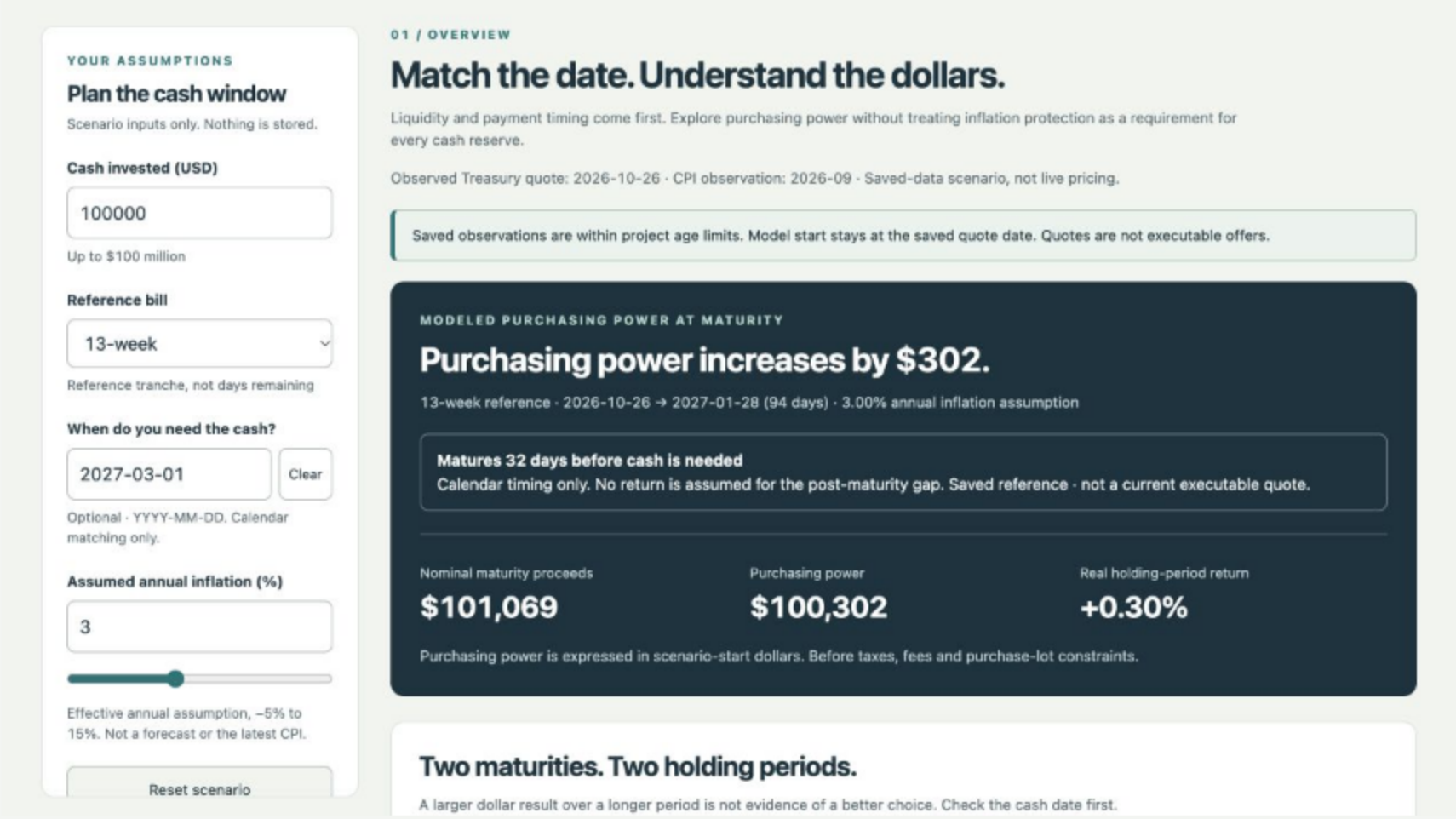
The client’s decision

Cash must be available when it is needed.

A fixed-dollar bill prioritizes timing and nominal safety.
A variable spending need also raises a purchasing-power question.

Enter a cash date; compare reference maturities and inflation.
Calendar matching does not establish settlement or cash availability.

The working dashboard



One holding period, two price changes

Observed discount quote and remaining days

Price per $100 = 100 × (1 − discount rate × days / 360)
Nominal proceeds = invested cash × 100 / price

Assumed inflation over the same days

Inflation factor = (1 + annual assumption)^(days / 365)
Purchasing power = nominal proceeds / inflation factor

Quote-date purchase, fractional holdings and no costs are simplifying assumptions.

A change in inflation changes purchasing power

$100,000 invested · 94 days · 4.05% bank discount quote

3% annual inflation assumption

$100,302

Modeled purchasing power

6% annual inflation assumption

$99,563

Modeled purchasing power

Nominal proceeds remain $101,069 in both scenarios.

Source: Treasury 2026-10-26 quote and stated scenario formulas. Amounts rounded.

Data and checks

Three observed series from original providers

Treasury: 13-week and 26-week bank discount quotes
BLS: all-items CPI-U, not seasonally adjusted
Snapshots retained with documented reuse terms

15 arithmetic/source checks + 62 revision logic checks
Actual HTTP desktop/mobile checks; file-mode opening pending

A student must still independently verify the results. Source dates: Treasury 2026-10-26; CPI 2026-09.

Supervised maintenance

Normal update: reveal a genuine withheld quote

23 October to 26 October 2026
Rates, reference maturity and modeled proceeds update together.

Failure trial: incompatible units

The process stops before writing a candidate.
Accepted files remain unchanged; release requires human review.

Revision maintenance rehearsals passed. Fresh-context v2 handover and human acceptance remain pending.

Handover and remaining decisions

The package preserves source inputs, formulas and review evidence.
The original version remains intact; this revision is a candidate.

Before submission

Supply the instructor’s identifiers.
Complete student checks and human accept/reject records.
Run the final extracted-file test and rehearse the live demo.

Calendar matching is included; settlement and receipt-of-funds modeling remain outside scope.